

# Jialong Fan

Third-Year Undergraduate, Physics + Computer Science, The Hong Kong University of Science and Technology |  
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## PhD Research Direction

Prospective PhD applicant broadly interested in quantum computing and its theoretical foundations, including quantum algorithms, quantum cryptography, fault-tolerant architectures, and mathematical structures connecting coding theory, information theory, and computation. Current preparation includes stabilizer/CSS codes, qLDPC/qLTC codes, high-dimensional expanders, decoder baselines, and simulation practice.

## Education

**The Hong Kong University of Science and Technology, Undergraduate Student in Physics + Computer Science** 2023 - Present

- Third-year undergraduate with cross-disciplinary training in physics and computer science.
- Academic interests lie at the interface of quantum theory, algorithms, secure information processing, mathematical structures, and computational methods.

## Research Preparation and Academic Experience

**Study in Quantum Error Correction and FTQC, Shilin Huang Group** Junior Year - Present

- Studying fault-tolerant quantum computation and quantum error correction topics with exposure to qLDPC codes, toric codes, BP-OSD decoders, and related variants.
- Developing a theoretical foundation for code constructions, decoding algorithms, and their roles in scalable fault-tolerant architectures.

**UROP Learning Experience in NV-Center Quantum Computing, Yang Sen Group, HKUST** Sophomore Fall

- Gained introductory exposure to NV-center-based quantum computing through an HKUST UROP project, building context for physical platforms and quantum information applications.

## Independent Study and Research Direction

- Independently studying high-dimensional expanders and their connections to quantum locally testable codes, with attention to locality, testability, distance, rate, and decodability.
- Particularly drawn to broad, mathematically structured problems in quantum computing, including secure protocols, algorithms, fault tolerance, and information-processing tasks.
- Comfortable learning across physics, mathematics, and computer science, and motivated to connect formal theory with simulation and algorithmic understanding.

## Technical Skills

|                           |                                                                                                        |
|---------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Programming</b>        | Python; basic C++; R.                                                                                  |
| <b>Quantum Simulation</b> | Experience using the Stim Python library for quantum-circuit and quantum-error-correction simulations. |
| <b>Relevant Topics</b>    | FTQC; qLDPC codes; toric codes; BP-OSD decoders and variants; high-dimensional expanders; qLTC.        |

## Language

TOEFL iBT: 113